

First Passage Time of a Discrete 1D Biased Random Walk:

A Rigorous, Step-by-Step Comparison of Two Absorbing vs. Mixed Boundaries

Abstract

This document provides a complete, mathematically rigorous, and exhaustively detailed derivation of the First Passage Time (FPT) distribution for a discrete 1D biased random walk. We explicitly construct and contrast two fundamental physical configurations: (1) a lattice bounded by two absorbing walls, and (2) a lattice bounded by a reflecting wall and an absorbing wall. Every algebraic manipulation, trigonometric identity, and linear algebra concept is derived from first principles.

1 Physical Setup and Walking Dynamics

Consider a particle performing a discrete-time random walk on a one-dimensional lattice. At each discrete time step t , the particle makes exactly one hop from its current site x :

- **Right hop:** Moves to $x + 1$ with probability p .
- **Left hop:** Moves to $x - 1$ with probability $q = 1 - p$.

We define the bias of the walk by the ratio p/q . If $p > q$, the walk has a rightward drift; if $p < q$, it has a leftward drift. The particle starts at site $x = 0$ at time $t = 0$, meaning the initial probability distribution is a Kronecker delta: $P_0(x) = \delta_{x,0}$.

We will analyze two distinct boundary configurations. In both cases, the valid, non-absorbing domain where the particle can exist is $x \in \{0, 1, \dots, N - 1\}$.

1.1 Case 1: Two Absorbing Boundaries

The system is bounded by absorbing walls at $x = -1$ and $x = N$.

Intuition 1. *Imagine a tightrope walker on a rope of length N . If they step off the left end ($x = -1$) or the right end ($x = N$), they fall and are permanently removed from the system.*

Mathematical Boundary Conditions:

$$P_t(-1) = 0 \quad \text{and} \quad P_t(N) = 0 \quad \text{for all } t \geq 0. \quad (1)$$

1.2 Case 2: Mixed Boundaries (Reflecting and Absorbing)

The system is bounded by a reflecting wall at $x = 0$ and an absorbing wall at $x = N$.

Intuition 2. *Imagine a corridor with a solid, bouncy wall on the left ($x = 0$) and a cliff on the right ($x = N$). If the walker tries to step left from $x = 0$, they bounce off the wall and remain at $x = 0$.*

Mathematical Boundary Conditions:

$$P_{t+1}(0) = qP_t(0) + qP_t(1) \quad (\text{Reflecting at left}), \quad (2)$$

$$P_t(N) = 0 \quad (\text{Absorbing at right}). \quad (3)$$

Note 1. *The reflecting condition at $x = 0$ states that the probability of being at $x = 0$ at time $t+1$ comes from two sources: (1) walkers already at $x = 0$ who attempted to hop left (probability q) and bounced back, and (2) walkers at $x = 1$ who hopped left (probability q).*

2 The Master Equations

Let $P_t(x)$ be the probability that the particle is at site x at time t and has *not yet been absorbed*. For any interior "bulk" site, the probability at time $t+1$ is replenished by arrivals from neighboring sites at time t .

2.1 The Bulk Equation (Shared by Both Cases)

For sites sufficiently far from the boundaries, the particle can arrive from the left (hopping right) or from the right (hopping left):

$$P_{t+1}(x) = pP_t(x-1) + qP_t(x+1). \quad (4)$$

2.2 Applying the Master Equation to the Boundaries

For Case 1 (Two Absorbing): Because $P_t(-1) = 0$ and $P_t(N) = 0$, the bulk equation (Eq. 4) actually applies perfectly to *every* valid site $x \in \{0, 1, \dots, N-1\}$ without modification. For example, at $x = 0$: $P_{t+1}(0) = pP_t(-1) + qP_t(1) = p(0) + qP_t(1) = qP_t(1)$. The absorbing boundary handles itself automatically.

For Case 2 (Mixed): The bulk equation applies to $x \in \{1, \dots, N-1\}$. However, at $x = 0$, we must explicitly use the reflecting boundary condition defined in Eq. 2:

$$P_{t+1}(0) = qP_t(0) + qP_t(1). \quad (5)$$

3 Solving the Master Equation: The Ansatz

Because the master equation is linear and the transition probabilities (p, q) do not change over time, the system is time-invariant. We seek solutions using the method of separation of variables. We assume the probability distribution can be factored into a purely spatial component and a purely temporal component:

$$P_t(x) = \phi(x)\mu^t. \quad (6)$$

Here, $\phi(x)$ is the spatial eigenfunction (the "shape" of the probability distribution), and μ is the temporal eigenvalue (the decay rate per time step).

3.1 Deriving the Spatial Eigenvalue Equation

Substitute the ansatz (Eq. 6) into the bulk master equation (Eq. 4):

$$\phi(x)\mu^{t+1} = p\phi(x-1)\mu^t + q\phi(x+1)\mu^t. \quad (7)$$

Factor out μ^t on the right side:

$$\phi(x)\mu^{t+1} = \mu^t [p\phi(x-1) + q\phi(x+1)]. \quad (8)$$

Divide both sides by μ^t (assuming $\mu \neq 0$):

$$\mu\phi(x) = p\phi(x-1) + q\phi(x+1). \quad (9)$$

This is a second-order linear homogeneous difference equation. It is the discrete spatial analog of the time-independent Schrödinger equation in quantum mechanics.

3.2 The Characteristic Equation

To solve linear difference equations with constant coefficients, we use the standard exponential ansatz for the spatial domain. Assume:

$$\phi(x) = \lambda^x, \quad (10)$$

where λ is a constant base. Substitute this into Eq. 9:

$$\mu\lambda^x = p\lambda^{x-1} + q\lambda^{x+1}. \quad (11)$$

To simplify, divide the entire equation by the lowest power of λ , which is λ^{x-1} :

$$\frac{\mu\lambda^x}{\lambda^{x-1}} = \frac{p\lambda^{x-1}}{\lambda^{x-1}} + \frac{q\lambda^{x+1}}{\lambda^{x-1}} \quad (12)$$

$$\mu\lambda = p + q\lambda^2. \quad (13)$$

Rearranging into standard quadratic form yields the characteristic equation:

$$q\lambda^2 - \mu\lambda + p = 0. \quad (14)$$

3.3 Roots of the Characteristic Equation

Using the quadratic formula, the two roots λ_+ and λ_- are:

$$\lambda_{\pm} = \frac{\mu \pm \sqrt{\mu^2 - 4pq}}{2q}. \quad (15)$$

Because this is a second-order equation, the general solution for the spatial eigenfunction is a linear combination of both roots:

$$\phi(x) = A\lambda_+^x + B\lambda_-^x, \quad (16)$$

where A and B are constants that will be determined by the boundary conditions.

3.4 Vieta's Formulas

From the properties of quadratic equations ($a\lambda^2 + b\lambda + c = 0$), we can immediately extract the product and sum of the roots without explicitly using the quadratic formula:

$$\text{Product: } \lambda_+ \lambda_- = \frac{c}{a} = \frac{p}{q}, \quad (17)$$

$$\text{Sum: } \lambda_+ + \lambda_- = -\frac{b}{a} = \frac{\mu}{q}. \quad (18)$$

4 Polar Parametrization of the Roots

4.1 The Need for Complex Numbers

Because the boundaries (whether absorbing or reflecting) force the probability to drop to zero or bounce back, the spatial eigenfunctions $\phi(x)$ must exhibit oscillatory, wave-like behavior. Furthermore, because probability is constantly leaking out of the absorbing boundary at $x = N$, the total probability must decay over time, meaning $|\mu| < 1$.

Let us examine the discriminant in the quadratic formula: $\Delta = \mu^2 - 4pq$. We know that $p + q = 1$. The maximum possible value of the product $4pq$ occurs when $p = q = 0.5$, yielding $4(0.5)(0.5) = 1$. Therefore, $4pq \leq 1$. Since we established that $|\mu| < 1$ for decaying modes, it follows that $\mu^2 < 1$. In the vast majority of cases, $\mu^2 < 4pq$, making the discriminant Δ strictly negative. A negative discriminant means the roots $\lambda_{\pm}$ are complex conjugates.

Intuition 3. *In linear algebra, real eigenvalues correspond to pure exponential growth or decay. Complex eigenvalues correspond to rotation and oscillation. Because the particle is "trapped" in a bounded domain, its probability wave must oscillate (to create nodes at the boundaries) while simultaneously decaying in amplitude (due to absorption). Complex roots perfectly capture this dual behavior.*

4.2 Transforming to Polar Coordinates

Working with the square root of a negative number in Cartesian form ($a \pm bi$) makes applying boundary conditions algebraically disastrous, especially when raising the roots to the power of N . Instead, we map the complex roots to polar coordinates.

Any complex number can be written as $re^{i\theta}$, where r is the magnitude (radius) and θ is the phase angle. Because the roots of a polynomial with real coefficients must be complex conjugates, we define:

$$\lambda_+ = re^{i\theta} \quad \text{and} \quad \lambda_- = re^{-i\theta}. \quad (19)$$

4.3 Finding the Magnitude r

We use Vieta's product formula (Eq. 17):

$$\lambda_+ \lambda_- = \frac{p}{q}. \quad (20)$$

Substitute the polar forms:

$$(re^{i\theta})(re^{-i\theta}) = r^2 e^{i\theta - i\theta} = r^2 e^0 = r^2. \quad (21)$$

Equating the two expressions for the product:

$$r^2 = \frac{p}{q} \implies r = \sqrt{\frac{p}{q}}. \quad (22)$$

Thus, the exact polar forms of the roots are:

$$\lambda_{\pm} = \sqrt{\frac{p}{q}} e^{\pm i\theta}. \quad (23)$$

4.4 Finding the Eigenvalue μ in terms of θ

We now use Vieta's sum formula (Eq. 18):

$$\lambda_+ + \lambda_- = \frac{\mu}{q} \implies \mu = q(\lambda_+ + \lambda_-). \quad (24)$$

Substitute the polar forms:

$$\mu = q \left(\sqrt{\frac{p}{q}} e^{i\theta} + \sqrt{\frac{p}{q}} e^{-i\theta} \right). \quad (25)$$

Factor out the magnitude $\sqrt{p/q}$:

$$\mu = q \sqrt{\frac{p}{q}} (e^{i\theta} + e^{-i\theta}). \quad (26)$$

Simplify the leading coefficient: $q \sqrt{p/q} = \sqrt{q^2 \cdot p/q} = \sqrt{pq}$.

$$\mu = \sqrt{pq} (e^{i\theta} + e^{-i\theta}). \quad (27)$$

Finally, apply Euler's trigonometric identity, which states that $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$:

$$\mu = 2\sqrt{pq} \cos \theta. \quad (28)$$

Intuition 4. *This is a profound result. The temporal decay rate μ of any mode is entirely governed by its spatial frequency θ . Because $|\cos \theta| \leq 1$ and $4pq \leq 1$, it is mathematically guaranteed that $|\mu| \leq 1$, ensuring the probability distribution decays or remains stable over time, perfectly matching the physical reality of an absorbing system.*

5 Applying Boundary Conditions and Quantization

In the previous section, we established that the spatial eigenfunctions take the general form $\phi(x) = A\lambda_+^x + B\lambda_-^x$, and that the roots can be elegantly parametrized in polar coordinates as $\lambda_{\pm} = re^{\pm i\theta}$, where the magnitude is $r = \sqrt{p/q}$ and the temporal eigenvalue is $\mu = 2\sqrt{pq} \cos \theta$.

The continuous parameter θ represents the spatial frequency of the probability wave. However, a physical system bounded by walls can only support specific, discrete resonant frequencies. We must now enforce the boundary conditions for both Case 1 and Case 2 to quantize θ into a discrete set of allowed values θ_n .

5.1 Case 1: Two Absorbing Boundaries

The physical domain is bounded by absorbing cliffs at $x = -1$ and $x = N$. This requires the spatial eigenfunction to vanish exactly at both boundaries:

$$\phi(-1) = 0 \quad \text{and} \quad \phi(N) = 0. \quad (29)$$

Step 1: Apply the left boundary condition ($x = -1$).

$$A\lambda_+^{-1} + B\lambda_-^{-1} = 0. \quad (30)$$

Multiply the entire equation by $\lambda_+\lambda_-$ to clear the negative exponents:

$$A\lambda_- + B\lambda_+ = 0 \implies \frac{A}{B} = -\frac{\lambda_+}{\lambda_-}. \quad (31)$$

Step 2: Apply the right boundary condition ($x = N$).

$$A\lambda_+^N + B\lambda_-^N = 0 \implies \frac{A}{B} = -\frac{\lambda_-^N}{\lambda_+^N} = -\left(\frac{\lambda_-}{\lambda_+}\right)^N. \quad (32)$$

Step 3: Equate the ratios to find the phase-matching condition. Since both expressions represent the same ratio A/B , we set them equal:

$$-\frac{\lambda_+}{\lambda_-} = -\left(\frac{\lambda_-}{\lambda_+}\right)^N. \quad (33)$$

Multiply both sides by $(\lambda_+/\lambda_-)^N$ to group the terms:

$$\left(\frac{\lambda_+}{\lambda_-}\right)^1 \cdot \left(\frac{\lambda_+}{\lambda_-}\right)^N = 1 \implies \left(\frac{\lambda_+}{\lambda_-}\right)^{N+1} = 1. \quad (34)$$

Step 4: Substitute the polar parametrization. Recall that $\lambda_+ = re^{i\theta}$ and $\lambda_- = re^{-i\theta}$. The ratio is:

$$\frac{\lambda_+}{\lambda_-} = \frac{re^{i\theta}}{re^{-i\theta}} = e^{2i\theta}. \quad (35)$$

Substitute this into the phase-matching condition:

$$(e^{2i\theta})^{N+1} = 1 \implies e^{2i(N+1)\theta} = 1. \quad (36)$$

By Euler's identity, $e^{i\Phi} = 1$ if and only if Φ is an integer multiple of 2π . Therefore:

$$2(N+1)\theta = 2\pi n \implies \theta_n = \frac{\pi n}{N+1}, \quad n = 1, 2, \dots, N. \quad (37)$$

Intuition 5. *This is the exact discrete analog of a guitar string pinned at both ends. The boundaries force the probability wave to have nodes (zeros) at $x = -1$ and $x = N$. The integer n represents the number of "half-wavelengths" that fit inside the domain. Because the boundaries are strictly zero, the quantization yields a beautifully simple, closed-form algebraic fraction.*

Step 5: Derive the exact eigenfunctions. Substitute the ratio $A/B = -e^{2i\theta_n}$ back into the general solution $\phi(x) = A\lambda_+^x + B\lambda_-^x$. Setting the arbitrary scale factor $B = 1$:

$$\phi_n(x) = -e^{2i\theta_n} (re^{i\theta_n})^x + (re^{-i\theta_n})^x = r^x [e^{-ix\theta_n} - e^{i(x+2)\theta_n}]. \quad (38)$$

To convert this difference of complex exponentials into a real sine function, we must factor out the "midpoint" phase to create symmetric exponents. The midpoint of $-x\theta_n$ and $(x+2)\theta_n$ is θ_n . Factoring out $e^{i\theta_n}$:

$$\phi_n(x) = r^x e^{i\theta_n} [e^{-i(x+1)\theta_n} - e^{i(x+1)\theta_n}]. \quad (39)$$

Apply Euler's sine identity, $e^{-iu} - e^{iu} = -2i \sin(u)$, with $u = (x+1)\theta_n$:

$$\phi_n(x) = r^x e^{i\theta_n} [-2i \sin((x+1)\theta_n)]. \quad (40)$$

Because eigenfunctions are defined only up to an arbitrary multiplicative constant, we drop the complex constant $-2ie^{i\theta_n}$. Substituting $r = \sqrt{p/q}$ and $\theta_n = \frac{\pi n}{N+1}$, we arrive at the exact spatial eigenfunctions for Case 1:

$$\phi_n(x) = \left(\frac{p}{q}\right)^{x/2} \sin\left(\frac{\pi n(x+1)}{N+1}\right). \quad (41)$$

5.2 Case 2: Mixed Boundaries (Reflecting and Absorbing)

The domain is bounded by an absorbing cliff at $x = N$ and a reflecting wall at $x = 0$.

$$\phi(N) = 0 \quad \text{and} \quad \mu\phi(0) = q\phi(0) + q\phi(1). \quad (42)$$

Step 1: Apply the absorbing boundary condition ($x = N$).

$$A\lambda_+^N + B\lambda_-^N = 0 \implies A = -B \left(\frac{\lambda_-}{\lambda_+}\right)^N. \quad (43)$$

Substitute this expression for A directly into the general solution $\phi(x) = A\lambda_+^x + B\lambda_-^x$:

$$\phi(x) = B \left[-\left(\frac{\lambda_-}{\lambda_+}\right)^N \lambda_+^x + \lambda_-^x \right]. \quad (44)$$

Factor out λ_-^N :

$$\phi(x) = B\lambda_-^N [-\lambda_+^{x-N} + \lambda_-^{x-N}]. \quad (45)$$

Now, substitute the polar forms $\lambda_{\pm} = re^{\pm i\theta}$. The prefactor $B\lambda_-^N$ is just a constant, so we drop it. We are left with the bracketed term:

$$\phi(x) \propto -(re^{i\theta})^{x-N} + (re^{-i\theta})^{x-N} = r^{x-N} [e^{-i(x-N)\theta} - e^{i(x-N)\theta}]. \quad (46)$$

Notice that $x - N = -(N - x)$. Substituting this into the exponents:

$$\phi(x) \propto r^x [e^{i(N-x)\theta} - e^{-i(N-x)\theta}]. \quad (47)$$

Apply Euler's sine identity, $e^{iu} - e^{-iu} = 2i \sin(u)$, with $u = (N - x)\theta$. Dropping the constant $2i$, we obtain the general functional form for the mixed boundary case:

$$\phi(x) = \left(\frac{p}{q}\right)^{x/2} \sin((N - x)\theta). \quad (48)$$

Step 2: Apply the reflecting boundary condition ($x = 0$). We must now force this wave to satisfy $\mu\phi(0) = q\phi(0) + q\phi(1)$. First, evaluate the necessary terms using our general form $\phi(x) = r^x \sin((N - x)\theta)$ and $\mu = 2qr \cos \theta$:

$$\phi(0) = r^0 \sin(N\theta) = \sin(N\theta), \quad (49)$$

$$\phi(1) = r^1 \sin((N - 1)\theta) = r \sin((N - 1)\theta). \quad (50)$$

Substitute these into the reflecting boundary equation:

$$(2qr \cos \theta) \sin(N\theta) = q \sin(N\theta) + q[r \sin((N - 1)\theta)]. \quad (51)$$

Divide the entire equation by q :

$$2r \cos \theta \sin(N\theta) = \sin(N\theta) + r \sin((N - 1)\theta). \quad (52)$$

Step 3: Trigonometric expansion and simplification. Apply the angle subtraction identity to $\sin((N - 1)\theta)$:

$$\sin((N - 1)\theta) = \sin(N\theta - \theta) = \sin(N\theta) \cos \theta - \cos(N\theta) \sin \theta. \quad (53)$$

Substitute this expansion back into the boundary equation:

$$2r \cos \theta \sin(N\theta) = \sin(N\theta) + r [\sin(N\theta) \cos \theta - \cos(N\theta) \sin \theta]. \quad (54)$$

Distribute the r on the right side:

$$2r \cos \theta \sin(N\theta) = \sin(N\theta) + r \sin(N\theta) \cos \theta - r \cos(N\theta) \sin \theta. \quad (55)$$

Subtract $r \sin(N\theta) \cos \theta$ from both sides to consolidate the terms containing r :

$$r \cos \theta \sin(N\theta) = \sin(N\theta) - r \cos(N\theta) \sin \theta. \quad (56)$$

Move the remaining r term to the left side:

$$r [\cos \theta \sin(N\theta) + \sin \theta \cos(N\theta)] = \sin(N\theta). \quad (57)$$

Recognize the term in the brackets as the angle addition identity for sine, $\sin(A + B) = \sin A \cos B + \cos A \sin B$, where $A = N\theta$ and $B = \theta$:

$$r \sin((N + 1)\theta) = \sin(N\theta). \quad (58)$$

Finally, substitute the magnitude $r = \sqrt{p/q}$ to obtain the exact **transcendental quantization condition** for Case 2:

$$\sin(N\theta_n) = \sqrt{\frac{p}{q}} \sin((N + 1)\theta_n). \quad (59)$$

Intuition 6. *Unlike Case 1, the reflecting wall does not force the probability wave to zero. Instead, the wave "bounces" off the wall. Because the random walk is biased ($p \neq q$), the bounce is asymmetric, introducing a complex phase shift. This phase shift prevents a simple algebraic fraction for θ_n . The allowed wavenumbers θ_n are the roots of this transcendental equation and must be found numerically. (Note: For the symmetric case $p = q = 1/2$, $r = 1$, and the equation reduces to $\sin(N\theta) = \sin((N + 1)\theta)$, which yields the closed-form roots $\theta_n = \frac{\pi(2n-1)}{2N+1}$).*

6 Full Eigenmode Expansion and Initial Conditions

Because the master equation is linear, any linear combination of valid eigenmodes is also a valid solution. The general surviving propagator is constructed via superposition:

$$P_t(x) = \sum_{n=1}^N c_n \phi_n(x) \mu_n^t. \quad (60)$$

To find the unknown expansion coefficients c_n , we enforce the initial condition at $t = 0$. Since $\mu_n^0 = 1$, the expansion must perfectly reconstruct the initial Dirac delta:

$$P_0(x) = \delta_{x,0} = \sum_{n=1}^N c_n \phi_n(x). \quad (61)$$

6.1 The Weighted Inner Product and Orthogonality

For a biased random walk ($p \neq q$), the transition matrix is non-symmetric. Consequently, the eigenfunctions are *not* orthogonal under the standard dot product. They are orthogonal only with respect to a specific weighted inner product. The correct weight function that restores orthogonality is $w(x) = (q/p)^x$.

We define the weighted inner product of two eigenfunctions as:

$$\langle \phi_m, \phi_n \rangle_w = \sum_{x=0}^{N-1} w(x) \phi_m(x) \phi_n(x) = \sum_{x=0}^{N-1} \left(\frac{q}{p}\right)^x \phi_m(x) \phi_n(x). \quad (62)$$

Notice what happens when we substitute the spatial bias term $\phi_n(x) \propto (p/q)^{x/2}$ into this inner product:

$$w(x) \phi_m(x) \phi_n(x) = \left(\frac{q}{p}\right)^x \left(\frac{p}{q}\right)^{x/2} \left(\frac{p}{q}\right)^{x/2} \times (\text{sine terms}) = \left(\frac{q}{p}\right)^x \left(\frac{p}{q}\right)^x \times (\text{sine terms}) = 1 \times (\text{sine terms}). \quad (63)$$

The spatial bias perfectly cancels the weight function! This mathematical miracle reduces the weighted inner product to a pure sum of sines, guaranteeing orthogonality:

$$\langle \phi_m, \phi_n \rangle_w = \mathcal{N}_n \delta_{mn}, \quad (64)$$

where $\mathcal{N}_n$ is the squared norm of the n -th eigenfunction, and δ_{mn} is the Kronecker delta.

To isolate a specific coefficient c_m , we take the weighted inner product of both sides of the initial condition equation with $\phi_m(x)$:

$$\sum_{x=0}^{N-1} w(x) \delta_{x,0} \phi_m(x) = \sum_{n=1}^N c_n \langle \phi_n, \phi_m \rangle_w. \quad (65)$$

Because of the Kronecker delta δ_{mn} on the right side, every term in the sum vanishes except when $n = m$. The right side collapses to $c_m \mathcal{N}_m$. On the left side, the Dirac delta $\delta_{x,0}$ ensures that only the $x = 0$ term survives the sum. Since $w(0) = (q/p)^0 = 1$, the left side simply evaluates to $\phi_m(0)$. Therefore:

$$\phi_m(0) = c_m \mathcal{N}_m \implies c_m = \frac{\phi_m(0)}{\mathcal{N}_m}. \quad (66)$$

6.2 Case 1: Coefficients for Two Absorbing Boundaries

For Case 1, the eigenfunctions are $\phi_n(x) = (p/q)^{x/2} \sin\left(\frac{\pi n(x+1)}{N+1}\right)$. The norm $\mathcal{N}_n$ is the standard discrete sine orthogonality sum. Letting $j = x + 1$:

$$\mathcal{N}_n = \sum_{j=1}^N \sin^2\left(\frac{\pi n j}{N+1}\right) = \frac{N+1}{2}. \quad (67)$$

Evaluating the eigenfunction at $x = 0$:

$$\phi_n(0) = \sin\left(\frac{\pi n}{N+1}\right). \quad (68)$$

Substituting these into our formula for c_n :

$$c_n = \frac{2}{N+1} \sin\left(\frac{\pi n}{N+1}\right). \quad (69)$$

6.3 Case 2: Coefficients for Mixed Boundaries

For Case 2, the eigenfunctions are $\phi_n(x) = (p/q)^{x/2} \sin((N-x)\theta_n)$. Evaluating at $x = 0$:

$$\phi_n(0) = \sin(N\theta_n). \quad (70)$$

The norm $\mathcal{N}_n$ requires summing $\sin^2((N-x)\theta_n)$. Letting $j = N - x$, the sum runs from $j = 1$ to N :

$$\mathcal{N}_n = \sum_{j=1}^N \sin^2(j\theta_n). \quad (71)$$

Using the half-angle identity $\sin^2(u) = \frac{1-\cos(2u)}{2}$ and the Lagrange trigonometric identity for the sum of cosines, this evaluates exactly to:

$$\mathcal{N}_n = \frac{N}{2} - \frac{\sin(N\theta_n) \cos((N+1)\theta_n)}{2 \sin(\theta_n)}. \quad (72)$$

Therefore, the expansion coefficients for the mixed boundary case are:

$$c_n = \frac{\sin(N\theta_n)}{\frac{N}{2} - \frac{\sin(N\theta_n) \cos((N+1)\theta_n)}{2 \sin(\theta_n)}}. \quad (73)$$

7 Survival Probability and First Passage Time

The survival probability S_t is defined as the total probability that the particle has not yet been absorbed by time t . It is the sum of the probabilities over all valid, non-absorbing sites:

$$S_t = \sum_{x=0}^{N-1} P_t(x). \quad (74)$$

The First Passage Time (FPT) distribution, F_t , is the probability that the particle is absorbed at *exactly* step t . By definition, this is the amount of survival probability lost between step $t - 1$ and step t :

$$F_t = S_{t-1} - S_t. \quad (75)$$

7.1 Case 1: Two Absorbing Boundaries

To find the recurrence relation for S_{t+1} , we sum the bulk master equation over all valid sites $x \in \{0, 1, \dots, N-1\}$:

$$S_{t+1} = \sum_{x=0}^{N-1} P_{t+1}(x) = p \sum_{x=0}^{N-1} P_t(x-1) + q \sum_{x=0}^{N-1} P_t(x+1). \quad (76)$$

We evaluate the two sums separately by re-indexing. **Right-hop sum:** Let $y = x - 1$. The limits change from $x \in [0, N-1]$ to $y \in [-1, N-2]$.

$$p \sum_{y=-1}^{N-2} P_t(y) = p \left[P_t(-1) + \sum_{y=0}^{N-2} P_t(y) \right]. \quad (77)$$

Because the left boundary is absorbing, $P_t(-1) = 0$. The remaining sum is the total survival probability minus the last site ($y = N-1$):

$$p \sum_{y=0}^{N-2} P_t(y) = p(S_t - P_t(N-1)). \quad (78)$$

Left-hop sum: Let $y = x + 1$. The limits change to $y \in [1, N]$.

$$q \sum_{y=1}^N P_t(y) = q \left[\sum_{y=1}^{N-1} P_t(y) + P_t(N) \right]. \quad (79)$$

Because the right boundary is absorbing, $P_t(N) = 0$. The remaining sum is the total survival probability minus the first site ($y = 0$):

$$q \sum_{y=1}^{N-1} P_t(y) = q(S_t - P_t(0)). \quad (80)$$

Adding these together and using $p + q = 1$, we obtain the survival recurrence:

$$S_{t+1} = S_t - pP_t(N-1) - qP_t(0). \quad (81)$$

Applying the definition $F_t = S_{t-1} - S_t$, the exact FPT distribution is the sum of the fluxes exiting *both* boundaries:

$$F_t = pP_{t-1}(N-1) + qP_{t-1}(0). \quad (82)$$

Closed-Form FPT for Case 1: Substitute the eigenmode expansion $P_{t-1}(x) = \sum c_n \phi_n(x) \mu_n^{t-1}$ into the FPT equation. Using the identities $\phi_n(0) = \sin(\frac{\pi n}{N+1})$ and $\phi_n(N-1) = \left(\frac{p}{q}\right)^{\frac{N-1}{2}} (-1)^{n+1} \sin(\frac{\pi n}{N+1})$, along with $c_n = \frac{2}{N+1} \sin(\frac{\pi n}{N+1})$, we obtain:

$$F_t = \frac{2}{N+1} \sum_{n=1}^N \sin^2\left(\frac{\pi n}{N+1}\right) \left[q + p(-1)^{n+1} \left(\frac{p}{q}\right)^{\frac{N-1}{2}} \right] \mu_n^{t-1}, \quad t \geq 1. \quad (83)$$

7.2 Case 2: Mixed Boundaries

For the mixed case, the master equation at $x = 0$ is explicitly $P_{t+1}(0) = qP_t(0) + qP_t(1)$. We separate this from the bulk sum:

$$S_{t+1} = P_{t+1}(0) + \sum_{x=1}^{N-1} P_{t+1}(x). \quad (84)$$

Applying the bulk master equation to the sum from $x = 1$ to $N - 1$ and re-indexing as before:

$$p \sum_{x=1}^{N-1} P_t(x-1) = p \sum_{y=0}^{N-2} P_t(y) = p(S_t - P_t(N-1)), \quad (85)$$

$$q \sum_{x=1}^{N-1} P_t(x+1) = q \sum_{y=2}^N P_t(y) = q(S_t - P_t(0) - P_t(1)) \quad (\text{since } P_t(N) = 0). \quad (86)$$

Now, assemble all pieces into S_{t+1} :

$$S_{t+1} = \underbrace{qP_t(0) + qP_t(1)}_{\text{from } x=0} + \underbrace{pS_t - pP_t(N-1)}_{\text{right-hops}} + \underbrace{qS_t - qP_t(0) - qP_t(1)}_{\text{left-hops}}. \quad (87)$$

Notice the mathematical miracle: the terms $qP_t(0)$ and $qP_t(1)$ perfectly cancel out!

$$S_{t+1} = (p+q)S_t - pP_t(N-1) = S_t - pP_t(N-1). \quad (88)$$

Consequently, the FPT is driven solely by the flux exiting the right cliff:

$$F_t = pP_{t-1}(N-1). \quad (89)$$

Intuition 7. *The algebraic cancellation is the mathematical signature of a true reflecting wall. The probability that attempts to leave the left boundary is instantly returned to the system. No probability leaks from the left; the only net loss is the flux hopping from $N-1$ to N .*

Closed-Form FPT for Case 2: Substitute the eigenmode expansion evaluated at $x = N - 1$. Using $\phi_n(N-1) = \left(\frac{p}{q}\right)^{\frac{N-1}{2}} \sin(\theta_n)$ and $c_n = \frac{\sin(N\theta_n)}{\mathcal{N}_n}$, we obtain:

$$F_t = p \left(\frac{p}{q}\right)^{\frac{N-1}{2}} \sum_{n=1}^N \frac{\sin(N\theta_n) \sin(\theta_n)}{\mathcal{N}_n} \mu_n^{t-1}, \quad t \geq 1. \quad (90)$$

8 Mean First Passage Time (MFPT)

While the MFPT $\langle T \rangle = \sum_{t=1}^{\infty} tF_t$ can be found by summing the infinite series, it is vastly more efficient to solve the backward Kolmogorov difference equation. Let T_x be the expected number of steps to reach *any* absorbing boundary starting from site x . The recurrence for the bulk is:

$$T_x = 1 + pT_{x+1} + qT_{x-1}. \quad (91)$$

The general solution to this non-homogeneous difference equation is the sum of the homogeneous solution and a particular linear solution $T_x = Cx$:

$$T_x = A + B \left(\frac{q}{p}\right)^x + \frac{x}{q-p}. \quad (92)$$

8.1 Case 1: Two Absorbing Boundaries

The boundary conditions are $T_{-1} = 0$ and $T_N = 0$.

$$T_{-1} = 0 \implies A + B \left(\frac{p}{q} \right) - \frac{1}{q-p} = 0, \quad (93)$$

$$T_N = 0 \implies A + B \left(\frac{q}{p} \right)^N + \frac{N}{q-p} = 0. \quad (94)$$

Subtracting the first equation from the second eliminates A and yields B :

$$B = \frac{-(N+1)}{(q-p) \left[\left(\frac{q}{p} \right)^N - \frac{p}{q} \right]}. \quad (95)$$

We seek the MFPT starting from the origin, $T_0 = A + B$. Substituting $A = \frac{1}{q-p} - B(p/q)$ into T_0 yields:

$$T_0 = \frac{1}{q-p} + B \left(1 - \frac{p}{q} \right) = \frac{1}{q-p} + B \left(\frac{q-p}{q} \right). \quad (96)$$

Substituting the expression for B , the $(q-p)$ terms cancel, leaving the exact closed-form MFPT:

$$\langle T \rangle = \frac{1}{q-p} - \frac{N+1}{q \left[\left(\frac{q}{p} \right)^N - \frac{p}{q} \right]}. \quad (97)$$

Intuition 8. *The first term represents the baseline drift. The second term is a massive correction factor representing the probability that the particle "falls off" the left cliff prematurely, drastically shortening its expected lifespan compared to an unbounded walk.*

8.2 Case 2: Mixed Boundaries

The boundary conditions are $T_N = 0$ and the reflecting condition at $x = 0$: $T_0 = 1 + pT_1 + qT_0 \implies T_0 - T_1 = 1/p$. Applying $T_N = 0$ gives $A = -B(q/p)^N - \frac{N}{q-p}$. Applying $T_0 - T_1 = 1/p$ to the general solution yields:

$$B \left(1 - \frac{q}{p} \right) - \frac{1}{q-p} = \frac{1}{p} \implies B \left(\frac{p-q}{p} \right) = \frac{q}{p(q-p)} \implies B = \frac{-q}{(p-q)^2}. \quad (98)$$

Evaluating $T_0 = A + B$ and substituting A and B :

$$\langle T \rangle = \frac{N}{p-q} - \frac{q}{(p-q)^2} \left[1 - \left(\frac{q}{p} \right)^N \right]. \quad (99)$$

Intuition 9. *The first term $\frac{N}{p-q}$ is the time it would take to drift distance N in an unbounded domain. The second term is the "boundary correction" caused by the reflecting wall. If the drift is to the right ($p > q$), the wall rarely affects the particle. If the drift is to the left ($p < q$), the particle is constantly smashed against the left wall, taking a massive amount of time to fight the drift and reach N .*

9 Comprehensive Comparison Table

The following table summarizes the profound mathematical and physical differences between the two boundary configurations derived in this document.

Table 1: Comparison of Two Absorbing vs. Mixed Boundary Configurations

Feature	Case 1: Two Absorbing Walls	Case 2: Mixed (Reflecting-Absorbing)
Physical Domain	Cliffs at $x = -1$ and $x = N$.	Hard wall at $x = 0$, cliff at $x = N$.
Boundary Conditions	$P_t(-1) = 0, \quad P_t(N) = 0$	$P_{t+1}(0) = qP_t(0) + qP_t(1),$ $P_t(N) = 0$
Quantization Condition	Closed-form: $\theta_n = \frac{\pi n}{N+1}$	Transcendental: $\sin(N\theta_n) = \sqrt{\frac{p}{q}} \sin((N+1)\theta_n)$
Spatial Eigenfunctions	$\left(\frac{p}{q}\right)^{x/2} \sin\left(\frac{\pi n(x+1)}{N+1}\right)$	$\left(\frac{p}{q}\right)^{x/2} \sin((N-x)\theta_n)$
Survival Recurrence	$S_{t+1} = S_t - p P_t(N-1) - q P_t(0)$	$S_{t+1} = S_t - p P_t(N-1)$
Probability Leakage	Leaks from <i>both</i> boundaries.	Leaks <i>only</i> from the right boundary.
First Passage Time (F_t)	$p P_{t-1}(N-1) + q P_{t-1}(0)$	$p P_{t-1}(N-1)$
Mean FPT (Biased)	$\frac{1}{q-p} - \frac{N+1}{q[(q/p)^N - p/q]}$	$\frac{N}{p-q} - \frac{q}{(p-q)^2} \left[1 - \left(\frac{q}{p}\right)^N\right]$
MFPT Physical Meaning	Shortened by premature left-cliff falls.	Lengthened by bouncing off the left wall against the drift.